Emily Sandford

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Research Interests

Exoplanet and brown dwarf interior modelling; exoplanet detection and characterisation with transit photometry; stellar and solar activity; and nonlinear dynamics and chaos, particularly phase space reconstruction of experimental observations.

Appointments

Leiden Observatory, Leiden University, Leiden, NL
- Postdoctoral Researcher — 01/06/2024 - present

Gonville & Caius College, University of Cambridge, Cambridge, UK
- Research Fellow — 01/10/2020 - 31/05/2024

Education

Columbia University, New York, NY
- Ph.D., Astronomy — 31/07/2020
  - Dissertation: "The Shapes of Planet Transits and Planetary Systems"
  - Supervisor: Prof. David Kipping
- M.A., M.Phil., Astronomy — 31/05/2016, 31/05/2017
  - Supervisors: Prof. David Kipping, Prof. Kathryn Johnston, Dr. Andreas Küpper

Yale University, New Haven, CT
- B.Sc., Physics, Cum Laude, with distinction in the Physics major — 31/05/2014
  - Supervisor: Prof. Marla Geha

Publications

17. I. Davis, J. Friske, E. Ilin, **E. Sandford** et al. The precarity of temporary astronomical staff in Dutch astronomy and how to address it, in prep.
16. **E. Sandford**. Attractor reconstruction for uncertain, unevenly sampled time series, in prep.
15. **E. Sandford** & Y. Miguel. Thermodynamically consistent equation of state tables for use in giant planet and brown dwarf interior modeling. Submitted to MNRAS, 2026.
14. A. Elias-López, A. Vidotto, D. Viganò, J. R. Fuentes, S. Müller, & **E. Sandford**. Estimations for strong and weak hot Jupiter magnetism: A Python package with interpolators and scaling laws. Submitted to A&A, 2026.
13. **E. Sandford**, F. Auchère, A. Mortier, L. Hayes, & D. Müller. Presenting 28 years of Sun-as-a-star extreme ultraviolet light curves from SOHO EIT. A&A, 2025, 699, 353.
12. M. Zilinskas, C. van Buchem, S. Zieba, Y. Miguel, **E. Sandford**, R. Hu, J. Patel, A. Bello-Arufe, L. Janssen, S.-M. Tsai, D. Dragomir, & M. Zhang. Characterising the atmosphere of 55 Cancri e: 1D forward model grid for current and future JWST observations. A&A, 2025, 697, 34.
11. **E. Sandford**, D. Kipping, & M. Collins. On Planetary Systems as Ordered Sequences. MNRAS, 2021, 505, 2224.
10. **E. Sandford**, D. Kipping, & M. Collins. The Multiplicity Distribution of *Kepler*'s Exoplanets. MNRAS, 2019, 489, 3162.
9. **E. Sandford**, N. Espinoza, R. Brahm, & A. Jordán. Estimation of Singly-Transiting K2 Planet Periods with Gaia Parallaxes. MNRAS, 2019, 489, 3149.
8. Z. Penoyre & **E. Sandford**. Higher Order Harmonics in the Light Curves of Eccentric Planetary Systems. MNRAS, 2019, 488, 4181.
7. **E. Sandford** & D. Kipping. Shadow Imaging of Transiting Objects. AJ, 2019, 157, 42.

6. D. Kipping, **E. Sandford**, & T. Jansen. Over 2000 *Kepler* Phase Curves from `Phasma`. RNAAS, 2018 2b, 14.
5. **E. Sandford** & D. Kipping. Know the Planet, Know the Star: Precise Stellar Densities from *Kepler* Transit Light Curves. AJ, 2017, 154, 288.
4. **E. Sandford**, A. H. W. Küpper, K. V. Johnston, & J. Diemand. Quantifying Tidal Stream Disruption in a Simulated Milky Way. MNRAS, 2017, 470, 522.
3. D. Kipping, C. Cameron, J. D. Hartman, J. R. A. Davenport, J. M. Matthews, D. Sasselov, J. Rowe, R. J. Siverd, J. Chen, **E. Sandford** et al. No Conclusive Evidence for Transits of Proxima b in MOST Photometry. AJ, 2017, 153, 93.
2. D. Kipping & **E. Sandford**. Observational Biases of Transiting Planets. MNRAS, 2016, 463, 1323.
1. D. Kipping, G. Torres, C. Henze, A. Teachey, H. Isaacson, E. Petigura, ... & **E. Sandford**. A Transiting Jupiter Analog. ApJ, 2016, 820, 112.

Invited Talks

14. 28 Years of Sun-as-a-Star Extreme Ultraviolet Light Curves from SOHO EIT. Solar-stellar and star-planet connections: mind the gap, Royal Astronomical Society, London, March 2026.
13. The climate costs and ethical perils of generative AI. Netherlands Astronomy Conference, May 2025.
12. Attractor reconstruction of active stellar light curves. University of Oxford, May 2024.
11. Attractor reconstruction of Solar quasi-periodicity. ESA ESTEC, March 2024.
10. How to picture quasi-periodic signals. ESA ESTEC, March 2024.
9. Attractor reconstruction of experimental time series. Imperial College London, November 2023.
8. Attractor reconstruction of active stellar light curves. Leiden University, February 2023.
7. Order or randomness in stellar light curves? Caius Science Network, Gonville & Caius College, University of Cambridge, November 2021.
6. On Planetary Systems as Ordered Sequences. Yale University, March 2021.
5. Computational Linguistics for Exoplanetary Systems. Machine Learning in Science & Engineering, Columbia U Data Science Institute, December 2020.
4. Shadow Imaging of Transiting Objects. University of California, Berkeley, August 2020.
3. Planetary Systems as Ordered Sequences. University of Cambridge, October 2019.
2. Shadow Imaging of Transiting Objects. Pennsylvania State University Center for Exoplanets and Habitable Worlds, March 2019.
1. Shadow Imaging of Transiting Objects. Pontifícia Universidad Católica de Chile, Santiago, Chile, March 2018.

Collaborations & outstanding proposals

Co-investigator, JWST cycle 6 proposals: Oct 2026
"The Glow of the Golden Microscope: AU Mic b in thermal emission", PI Z. Berta-Thompson
"Beyond the Permanent Dayside: A Warm Jupiter at the Tidal-Locking Transition", PI K. Duer-Milner
"Closing the 18-Micron Atmospheric Gap for 55 Cancri e with MIRI", PI S. Zieba

Planetary EOS Archive and Reference Library (PEARL) collaboration August 2026 - present

PLATO working group WP116100, Gas & Ice Giant Composition (chair T. Guillot) Oct 2025 - present

PLATO consortium member Oct 2025 - present

Teaching and Advising

Supervisor, MSc/BSc projects:

- Maria Garro, 1st-year MSc, Leiden University, "Latitudinally dependent Sun-as-a-star extreme ultraviolet light curves from SOHO/EIT" 2026-present
- Viesturs Strelcs (co-supervised with K. Duer-Milner), 1st-year MSc, Leiden University, "Time-evolving gas giant atmospheric dynamics" 2026-present
- Hakimeh Moini, MSc, Alzahra University, "Identifying EUV diagnostics in optical Sun-as-a-star spectra" 2026-present
- Emma Volkers (co-supervised with S. Ulmer-Moll and E. Van Dijk), 1st-year MSc, Leiden University, "Comparing planetary evolution models" 2025-2026
- Jip Wernke, final-year BSc, Leiden University, "Evolving Jupiter in MESA" 2025-2026

Supervisor, Ethiopian Space Science Society Student Research Program September 2026 - present
Students advised (with S. Platymesi): Kena Amenu, Felix Kaya Mambo, Dechasa Birhanu Hatau, Cherono Elizabeth, "Reanalyzing the TRAPPIST-1 system"

Supervisor, Leiden/ESA Astrophysics Program for Summer Students June - August 2026
Students advised (with A. Elias-López): Marion Asasira, "Attractor reconstruction for active Kepler stars"

Co-leader, Leiden Observatory discussion series on generative AI use in academia, Oct - Dec 2025
"Considerations for generative AI use in research"
"Against generative AI in education" (with Marcel van Daalen)

Lecturer, Cambridge M.Phil. in Data-Intensive Science April - June 2024
"Applications of Data Science to Exoplanets"

Guest lecturer, Cambridge Digital Humanities workshop on LLMs at Work December 2023

Research mentor, Nuffield Research Placement August - October 2023
Students advised: Adam Clarke, "Simulating sunspot light curves"

Guest lecturer, Gonville & Caius College postgraduate seminar series November 2022
"How to Teach in a Climate Crisis"

Supervisor, University of Cambridge Part II Astrophysics
Astrophysical Fluid Dynamics January 2022 - May 2023
Structure and Evolution of Stars October 2021 - May 2022

Guest lecturer, Astrostatistics, Columbia University April 2021
"A Superficial Introduction to Neural Networks"

Research mentor, American Museum of Natural History Science Research Mentoring Program
September 2017 - May 2018
Students advised: Christopher Ambrus, Catherine Atalig, James Hamue, and Caroline Klewinowski, "The Kepler Atlas, an interactive 3D model of *Kepler*'s exoplanet discoveries"

Instructor: Astronomy Lab I, Astronomy Lab II September 2015 - May 2017

Teaching assistant: Astrostatistics, Modeling the Universe, Another Earth September 2015 - May 2016

Successful Proposals

ESA Archival Research Visitor Programme, cycle 5 awarded June 2023
"Multi-wavelength attractor reconstruction of the Solar rotation cycle"
For 2-month visit to ESTEC to work on archival data from the Solar and Heliospheric Observatory, February - March 2024

Awards/Prizes

Shortlisted for the Caroline Herschel Prize Lectureship June 2023

Columbia University President's Global Innovation Fund Grant, for study in Santiago, Chile Mar 2018

Columbia University Dean's Fellowship August 2014 - August 2020

Honorable Mention, National Science Foundation Graduate Research Fellowship March 2016

American Astronomical Society Chambliss Student Poster Award January 2014

Open-Source Code Development

Lead developer:

- `spotchaos`, a `Python` package for attractor reconstruction — ongoing
- `EightBitTransit`, a `Python` package which generates light curves of arbitrary transiting shapes, and infers the transiting shape which produced an arbitrary light curve. — April 2019

Contributing developer:

- `single`, a `Python` package which fits single transits using stellar density information. — Nov 2019
- `OoT`, a `Python` package which generates self-consistent planet light curves including transits, secondary eclipses, tides, reflections, and relativistic beaming. — Sept 2019
- `SEDBuilder`, a `Python` package which collates archival photometric data points for any object with a 2MASS ID and generates its SED. — Sept 2018
- Kepler Atlas, a `javascript`-implemented interactive 3D model of *Kepler* planets. — May 2018

Public Outreach

Cool Worlds YouTube channel contributor — March 2016 - July 2021
Cambridge Institute of Astronomy public outreach talk, "Oh, the Planets You'll Go!" — February 2023
Cambridge Creative Encounters collaboration with artist Dr. Alina Loth — April - September 2020
Cambridge Behind the Curtains workshop with local playwrights — May - June 2020
Sky & Telescope freelance contributor — August - November 2018
Astrobites staff writer — January 2017 - October 2018

Organizing & Service

Netherlands Temporary Astronomy Staff Committee founding member — October 2025 - present
Leiden Observatory postdoc representative — September 2025 - present
Leiden Observatory Institute Council member — September 2025 - present
Leiden Obseratory LGBTQ+-in-astro coffee chat organizer — September 2024 - present
Co-organizer and panel member for Netherlands Astronomy Conference splinter session "A future for astronomers in the Netherlands: Solving the precarity crisis" — May 2026
Co-author, Open Letter: Stop the Uncritical Adoption of AI Technologies in Academia — June 2025
Justice for College Supervisors negotiating committee member — March 2023 - May 2024
Anti-casualisation rep. on the Cambridge UCU Executive Committee — September 2023 - May 2024
Cambridge LGBTQ+-in-astro coffee chat organizer — August 2023 - May 2024
Cambridge exoplanet seminar committee chair — June 2022 - May 2024
Cambridge postdoc committee member — September 2022 - May 2024
Leverhulme Centre for Life in the Universe Annual Meeting organizer — December 2022 - March 2023
Cambridge exoplanets group meeting organizer — October 2021 - June 2022
Astrobites editorial committee chair & vision committee chair — September 2018 - September 2019
Columbia Astronomy graduate student representative to faculty meetings — September 2017 - May 2019
Columbia Astrophysics Laboratory computing committee member — September 2017 - May 2018
Mentor, Columbia Astronomy graduate mentorship program — August 2016 - August 2020
Columbia Astronomy graduate admissions committee member — December 2016 - March 2017